

Should Platforms Be Allowed to Sell on Their Own Marketplaces?

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PART 1

Research Question

Presenter: Jinghao Mu

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Research Question

The paper asks three linked questions.

1. Does **banning** dual mode always **raise consumer surplus and welfare**?
2. If dual mode is banned, does the platform **switch to seller mode or marketplace mode**?
3. Are targeted **rules on imitation and self-preferring** better than a full ban?

Key Insight

The problem is not dual mode itself, but how the platform uses its power.

- A ban on dual mode is not always good for consumers or welfare.
- Rules that target imitation and self-preferring can work better.

Location and Contribution

The paper is located at the intersection of platform economics and antitrust policy.

- It studies a platform that can choose among **marketplace mode**, **seller mode**, and **dual mode**.
- It connects the policy debate on platform bans with a formal welfare analysis.
- It separates **dual mode itself** from two possible bad conducts: **product imitation** and **self-preferring**.

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PART 2

Baseline Model

Presenter: Jialong Zheng

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Players I: Consumers and Competitive Benchmark

Consumers

- Outside option of not buying anything (e.g., postpone purchase or buy another product): $v_o \sim G^a$
- Platform transaction benefit: $b > 0$ (finding products, one-click payment, delivery tracking, easy returns).
- Choice rule: buy the option with the highest net utility.

^a G is log-concave and continuously differentiable on $[0, \bar{v}_o]$, with density g .

Fringe sellers

- At least two identical sellers.
- Sell a basic product with value v .
- Consumers already know them and can buy directly.
- Bertrand competition gives the price benchmark.^a

^aAll marginal costs are normalized to zero.

Values and Parameters

Product values are:

Product	Consumer value	Meaning
Fringe sellers	ν	Basic product
Innovative seller S	$\nu + \Delta$	Better product
Platform M	$\nu + \sigma$	Platform's own product

- $b > 0$: convenience benefit from buying through the platform.
- $\nu_o \sim G$: outside option.
- τ : platform commission.

Intuition

$\sigma > 0$ means the platform is a better seller than fringe sellers; $\sigma < 0$ means the opposite.



Product Discovery

Consumers do not know about S at the start.

- Consumers know the fringe sellers.
- Consumers may know the platform's own product.
- Consumers discover S only if S is available on the platform.

Why this matters

The platform is not only a sales channel. It is also a discovery channel.

Intuition

If a policy changes the platform's mode, it may also change whether consumers find S .

Three Platform Modes

Mode	What M does	Key feature
Marketplace	Hosts third-party sellers	Earns commission τ
Seller	Sells its own product	No marketplace for S
Dual	Hosts sellers and sells own product	Both roles at once

Intuition

The policy debate is about whether the third mode should be allowed.



Timing

The game is solved by backward induction.

1. M chooses its mode and sets commission τ if there is a marketplace.
2. S decides whether to join the platform and chooses Δ .
3. Sellers set prices.
4. Consumers choose what to buy.

Equilibrium concept

Subgame Perfect Nash Equilibrium.



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PART 3

Equilibrium Analysis

Presenter: Binghui Guo

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Marketplace Equilibrium

Proposition 1 follows from two binding constraints. Since S **sells through the platform** iff $\tau \leq b$, M faces the *showrooming constraint* when choosing its commission.

Showrooming constraint

The platform fee binds at

$$\tau^{mkt} = b$$

M charges the highest fee that still keeps trade on the platform.

Competitive constraint

The inside price binds at

$$p_i^* = \tau^{mkt} + \Delta^{mkt} = b + \Delta^{mkt}$$

Hence demand is

$$G(v + b + \Delta^{mkt} - p_i^*) = G(v).$$

$$K'(\Delta^{mkt}) = G(v), \quad \Pi^{mkt} = bG(v), \quad \pi^{mkt} = \Delta^{mkt}G(v) - K(\Delta^{mkt}).$$

Intuition

S sells exclusively through the marketplace; innovation is chosen until marginal innovation cost equals the induced transaction volume.

Seller Mode: Setup

In seller mode, M sells only its own product. Seller S is unavailable on the platform, so consumers never discover S through M .

Seller S

- Its outside price is payoff-irrelevant.
- It chooses the lowest innovation level:

$$\Delta^{sell} = \Delta^l.$$

- It makes no sales in equilibrium.

Platform M

M chooses the price of its own product:

$$\max_{p_m \leq b + \sigma} p_m G(\nu + b + \sigma - p_m) \quad (3.1)$$

Price cap

The cap $p_m \leq b + \sigma$ comes from competition with fringe sellers in the direct channel: $\nu + \sigma + b - p_m \geq \nu$.

Seller Mode Equilibrium

The *competitive constraint* binds in the seller-mode pricing problem, so M charges the highest price that still beats fringe sellers.

Binding price

$$p_m^* = b + \sigma$$

The price equals convenience value plus M 's product advantage.

Demand and profits

Substituting p_m^* gives demand

$$G(\nu + b + \sigma - p_m^*) = G(\nu).$$

Therefore

$$\Pi^{sell} = (b + \sigma)G(\nu), \quad \pi^{sell} = 0.$$

Proposition 2

M sets $p_m^* = b + \sigma$ and sells to all consumers, whereas S sells to no one.

Intuition

Seller mode eliminates on-platform competition from S : the platform captures convenience value b plus its own product advantage σ .



Dual Mode: Setup

In dual mode, M is both an intermediary and a seller: it hosts S , charges commission τ , sells its own product, and sets its own price p_m .

Platform M

- sets commission τ ;
- sells its own product at price p_m ;
- earns either commission income or own-product revenue.

Seller S

- chooses innovation Δ ;
- chooses inside price p_i and outside price p_o ;
- competes with both fringe sellers and M 's own product.

New force

M 's own price becomes an extra cap on S 's inside price. This is the source of the price-squeeze mechanism.

Dual Mode: Pricing Subgame

Given τ and Δ , the stage-3 pricing outcome depends on whether M 's own product or S 's product has the higher quality.

Semi-seller equilibrium

If $\sigma \geq \Delta$, M 's product is weakly better:

$$p_i^* = \tau, \quad p_m^* = \tau + \sigma - \Delta$$

Consumers buy from M ; S earns zero operating margin.

Price-squeeze equilibrium

If $\Delta > \sigma$, S 's product is better:

$$p_i^* = p_m^* + \Delta - \sigma$$

Consumers buy from S , but M 's own price caps p_i^* .

Intuition

Dual mode matters when S is better: M may not sell, yet its own product disciplines S 's price.

Price Squeeze

When $\Delta > \sigma$, consumers buy from S through the marketplace, but M 's own price caps S 's inside price.

Feasible own-product price

The price-squeeze interval is

$$p_m^* \in [\max\{\tau - \Delta + \sigma, 0\}, \tau + \min\{\sigma, 0\}]$$

and the selected equilibrium uses the lowest feasible p_m^* .

Platform profit

Given $p_i^* = p_m^* + \Delta - \sigma$, demand is $G(\nu + \sigma + b - p_m^*)$.

Hence

$$\Pi = \tau G(\nu + \sigma + b - p_m^*)$$

Intuition

A lower p_m^* lowers S 's feasible price, expands demand, and raises commission income. This is why the selection rule picks $p_m^* = \max\{\tau - \Delta + \sigma, 0\}$.

Stage-2 Innovation Decision

After selecting the lowest feasible p_m^* , S 's payoff can be written as

$$\pi(\Delta) = \max\{(\Delta - \sigma - \tau)G(\nu + \sigma + b), 0\} - K(\Delta)$$

Threshold commission

Denote $\bar{\tau} \in (\Delta^l - \sigma, \bar{\Delta} - \sigma)$ by
 $(\bar{\Delta} - \sigma - \bar{\tau})G(\nu + b + \sigma) - K(\bar{\Delta}) = 0.$

Lemma 1

- If $\tau \leq \bar{\tau}$, then $\Delta = \bar{\Delta}$.
- If $\tau > \bar{\tau}$, then $\Delta = \Delta^l$.

Intuition

$\bar{\tau}$ is the highest commission compatible with high innovation; above it, price squeeze removes S 's incentive to pay $K(\bar{\Delta})$.

Stage-1 Commission Decision

In stage 1, M chooses τ while anticipating the innovation response. Two constraints shape the choice.

Low fee, high innovation

Set $\tau \leq \bar{\tau}$, induce $\Delta = \bar{\Delta}$, and earn commission on high demand:

$$\tau G(\nu + \sigma + b).$$

At the best such policy, $\tau = \min\{b, \bar{\tau}\}$.

High fee, low innovation

Set the showrooming upper bound $\tau = b$, accept $\Delta = \Delta^l$, and earn

$$(b + \max\{\sigma - \Delta^l, 0\})G(\nu + \Delta^l).$$

Where the constraints come from

The *showrooming constraint* $\tau \leq b$ keeps transactions on the platform; the *innovation constraint* $\tau \leq \bar{\tau}$ keeps S willing to choose high innovation.

Dual Mode Equilibrium

Proposition 3 gives M 's optimal commission after anticipating S 's innovation response.

- **Case 1:** $b \leq \bar{\tau}$. The showrooming upper bound already satisfies the innovation constraint:

$$\tau^{dual} = b, \quad \Delta^{dual} = \bar{\Delta}, \quad \Pi^{dual} = bG(\nu + \sigma + b).$$

- **Case 2:** $b > \bar{\tau}$. M cannot set the high fee b and preserve high innovation. It compares

$$\bar{\tau}G(\nu + \sigma + b) \quad \text{and} \quad (b + \max\{\sigma - \Delta^l, 0\})G(\nu + \Delta^l).$$

If the first term is larger, M sets $(\tau, \Delta) = (\bar{\tau}, \bar{\Delta})$; otherwise it sets $(\tau, \Delta) = (b, \Delta^l)$.

- **Trade-off in Case 2.** A lower fee preserves innovation and expands transactions; a higher fee extracts more per transaction but may destroy innovation.



Partial and Complete Price Squeeze

Price squeeze can be partial or complete depending on whether S keeps a positive operating margin after M 's selected price.

- **Partial price squeeze.** With high innovation,

$$\Delta^{dual} = \bar{\Delta}, \quad p_m^* = 0, \quad p_i^* = \bar{\Delta} - \sigma.$$

If $p_i^* > \tau^{dual}$, then S still earns positive margin:

$$p_i^* - \tau^{dual} > 0.$$

- **Complete price squeeze.** With low innovation and $\Delta^l > \sigma$, S still sells, but M sets p_m^* so that

$$p_i^* = \tau^{dual}.$$

Then S sells with zero operating margin.

- **Boundary case.** If $\Delta^l \leq \sigma$, M 's product is weakly better and the pricing subgame becomes the semi-seller equilibrium.

Which Mode Does the Platform Choose?

Corollary 1 compares platform profits across the three operating modes.

Pure modes

$$\Pi^{mkt} = bG(\nu), \quad \Pi^{sell} = (b + \sigma)G(\nu).$$

Therefore

$$\Pi^{mkt} \geq \Pi^{sell} \iff \sigma \leq 0.$$

Full choice

- Dual mode strictly dominates marketplace mode:
 $\Pi^{dual} > \Pi^{mkt}$.
- There is a unique $\underline{\sigma} > 0$ such that
 $\Pi^{dual} > \Pi^{sell} \iff \sigma < \underline{\sigma}$.
- Therefore:

$$M \text{ chooses } \begin{cases} \text{dual mode,} & \sigma < \underline{\sigma}, \\ \text{seller mode,} & \sigma \geq \underline{\sigma}. \end{cases}$$

Intuition

Thus M chooses dual mode when $\sigma < \underline{\sigma}$ and seller mode when $\sigma \geq \underline{\sigma}$. For small σ , price squeeze creates more transactions; for large σ , seller mode lets M exploit its own product advantage without revealing S .

Baseline Ban on Dual Mode: Mode Switch

Proposition 4 asks what happens if regulation removes dual mode from M 's choice set.

- **Post-ban choice set:**

$$\{\text{marketplace, seller, dual}\} \rightarrow \{\text{marketplace, seller}\}.$$

- **Post-ban mode choice:**

M chooses seller mode iff $\sigma > 0$, M chooses marketplace mode iff $\sigma \leq 0$.

- **Compared with the unconstrained baseline:**

$\sigma \geq \underline{\sigma}$:	seller $\rightarrow$ seller,	no real effect,
$0 < \sigma < \underline{\sigma}$:	dual $\rightarrow$ seller,	S disappears from the platform,
$\sigma \leq 0$:	dual $\rightarrow$ marketplace,	S remains but price squeeze disappears.



Baseline Ban on Dual Mode: Effects

Region	Mode switch	Π	π	CS	Δ	Welfare logic
$\sigma \geq \underline{\sigma}$	seller $\rightarrow$ seller	=	=	=	=	Ban is nonbinding because M already prefers seller mode.
$0 < \sigma < \underline{\sigma}$	dual $\rightarrow$ seller	$\downarrow$	$\downarrow$ or =	$\downarrow$	$\downarrow$ or =	Welfare falls: the ban removes S from consumer discovery and weakens transactions.
$\sigma \leq 0$	dual $\rightarrow$ marketplace	$\downarrow$	$\uparrow$	$\downarrow$	$\downarrow$ or $\uparrow$	Welfare is ambiguous; it rises only if innovation gains are large enough.

Key implication

A dual-mode ban always weakly lowers consumer surplus. It improves welfare only in the $\sigma \leq 0$ region and only when marketplace innovation gains dominate the loss of price-squeeze-driven transaction surplus.

When Can a Dual-Mode Ban Raise Welfare?

In the $\sigma \leq 0$ region, the ban changes the market from dual mode to marketplace mode. Welfare can rise only if two conditions hold.

- **Dual mode must have low innovation:**

$$\Delta^{dual} = \Delta^l.$$

Otherwise the ban replaces high dual-mode innovation $\bar{\Delta}$ with lower marketplace innovation Δ^{mkt} .

- **Marketplace innovation gains must be large enough:**

$$(b + \Delta^{mkt})G(\nu) - K(\Delta^{mkt}) > bG(\nu + \Delta^l) + \int_0^{\nu + \Delta^l} [\nu + \Delta^l - \max\{\nu, \nu_o\}] dG(\nu_o)$$

Interpretation

The left side is marketplace value from platform convenience plus higher innovation. The right side is the dual-mode low-innovation welfare benchmark.

Derivation: Marketplace Price Bounds

On-platform price bound

Fringe sellers on M provide utility $v + b - \tau$. If consumers buy S on M , utility is $v + \Delta + b - p_i$. Hence S must satisfy

$$v + \Delta + b - p_i \geq v + b - \tau \quad \Rightarrow \quad p_i \leq \tau + \Delta.$$

Direct-channel price bound

If consumers buy directly from S , utility is $v + \Delta - p_o$. Comparing this with fringe sellers on M gives

$$v + \Delta - p_o \geq v + b - \tau \quad \Rightarrow \quad p_o \leq \tau + \Delta - b.$$

[Back to price bounds](#)



Derivation: Showrooming Constraint

Compare S 's two channels

Under the binding price bounds, both channels induce the same transaction utility, $v + b - \tau$, but they give S different margins:

$$\text{through } M : (\tau + \Delta) - \tau = \Delta, \quad \text{directly : } \tau + \Delta - b = \Delta + \tau - b.$$

Channel choice

S sells through M iff

$$\Delta \geq \Delta + \tau - b \iff \tau \leq b.$$

Therefore any $\tau > b$ makes S use M for discovery but shift transactions to the direct channel.

Back to showrooming formula

Derivation: Marketplace Competitive Constraint

For inside sales, S solves

$$\max_{p_i \leq \tau + \Delta} (p_i - \tau) G(\nu + b + \Delta - p_i).$$

Why the upper bound binds

The derivative at the competitive upper bound $p_i = \tau + \Delta$ is

$$G(\nu + b - \tau) - \Delta g(\nu + b - \tau).$$

Since $\tau \leq b$ and G/g is increasing under log-concavity of G , Assumption (2) implies

$$\Delta \leq \bar{\Delta} < \frac{G(\nu)}{g(\nu)} \leq \frac{G(\nu + b - \tau)}{g(\nu + b - \tau)}.$$

The derivative is positive at the bound, so S would like to raise price further; the *competitive constraint* stops it.

[Back to competitive formula](#)

Derivation: Seller Competitive Constraint

In seller mode, M solves

$$\max_{p_m \leq b + \sigma} p_m G(v + b + \sigma - p_m).$$

Why the upper bound binds

At the competitive upper bound $p_m = b + \sigma$, the derivative is

$$G(v) - (b + \sigma)g(v).$$

Assumption (2) gives

$$b + \sigma < \frac{G(v)}{g(v)},$$

so the derivative is positive at the bound. Therefore M would like to raise price further, but the direct-channel fringe option imposes the *competitive constraint*.

[Back to seller equilibrium](#)

Derivation: Semi-Seller Pricing

Why S prices at effective marginal cost

When $\sigma \geq \Delta$, M 's own product is weakly better. Seller S pays commission τ on each platform sale, so its effective marginal cost is τ :

$$p_i^* = \tau.$$

Why M sells at $p_m^* = \tau + \sigma - \Delta$

Consumers are indifferent between M and S when

$$v + \sigma + b - p_m^* = v + \Delta + b - p_i^*.$$

Substituting $p_i^* = \tau$ gives

$$p_m^* = \tau + \sigma - \Delta.$$

[Back to pricing cases](#)



Derivation: Price-Squeeze Price Cap

Price cap imposed by M

Consumers buy S rather than M only if

$$v + \Delta + b - p_i \geq v + \sigma + b - p_m, \quad \Rightarrow \quad p_i \leq p_m + \Delta - \sigma.$$

The competitive constraint binds, so $p_i^* = p_m^* + \Delta - \sigma$.

Why this is price squeeze

M may not sell in equilibrium, but a lower p_m^* tightens the upper bound on S 's price and forces S to charge less.

[Back to price squeeze relation](#)



Derivation: Feasible Interval for p_m^*

Upper bounds

- $p_m^* \leq \tau$: otherwise M could undercut S and earn an own-product margin above the commission τ .
- $p_m^* \leq \tau + \sigma$: otherwise consumers prefer platform fringe sellers, since

$$v + \sigma + b - p_m^* < v + b - \tau.$$

Hence $p_m^* \leq \tau + \min\{\sigma, 0\}$.

[Back to feasible interval](#)

Lower bounds

- S must not sell below effective marginal cost: $p_i^* \geq \tau$.
- Since $p_i^* = p_m^* + \Delta - \sigma$, $p_m^* \geq \tau - \Delta + \sigma$.
- M 's own price is nonnegative: $p_m^* \geq 0$.

Hence $p_m^* \geq \max\{\tau - \Delta + \sigma, 0\}$.

$$p_m^* \in [\max\{\tau - \Delta + \sigma, 0\}, \tau + \min\{\sigma, 0\}].$$



Derivation: Selected Price in Price Squeeze

In a price-squeeze equilibrium, consumers buy from S and M earns commission revenue:

$$\Pi = \tau G(\nu + \sigma + b - p_m^*).$$

Selection rule

Because $G(\cdot)$ is increasing, a lower p_m^* raises transaction volume and therefore raises M 's commission profit. The selected equilibrium sets

$$p_m^* = \max\{\tau - \Delta + \sigma, 0\}.$$

[Back to selected price](#)



Derivation: Platform Profit in Price Squeeze

In the price-squeeze equilibrium, consumers buy S through M , so the platform earns commission τ per transaction.

Demand under price squeeze

Using $p_i^* = p_m^* + \Delta - \sigma$,

$$\nu + \Delta + b - p_i^* = \nu + \Delta + b - (p_m^* + \Delta - \sigma) = \nu + \sigma + b - p_m^*.$$

Demand is therefore $G(\nu + \sigma + b - p_m^*)$.

Commission profit

$$\Pi = \tau G(\nu + \sigma + b - p_m^*).$$

Since $G(\cdot)$ is increasing, the selected lowest p_m^* maximizes M 's commission profit among price-squeeze equilibria.

[Back to platform profit](#)

Derivation: Stage-2 Innovation Decision

With the selected lowest p_m^* , S 's payoff combines the zero-margin semi-seller case and the positive-margin price-squeeze case:

$$\pi(\Delta) = \max\{(\Delta - \sigma - \tau)G(\nu + \sigma + b), 0\} - K(\Delta).$$

Definition of the innovation threshold

Lemma 1 denotes $\bar{\tau} \in (\Delta^l - \sigma, \bar{\Delta} - \sigma)$ as the commission that makes S indifferent between high innovation and the zero-profit low-innovation option:

$$(\bar{\Delta} - \sigma - \bar{\tau})G(\nu + b + \sigma) - K(\bar{\Delta}) = 0.$$

Equivalently,

$$\bar{\tau} = \bar{\Delta} - \sigma - \frac{K(\bar{\Delta})}{G(\nu + b + \sigma)}.$$

Back to seller payoff

Back to innovation threshold



Derivation: Proposition 3 Profit Comparison

If $b \leq \bar{\tau}$, the showrooming upper bound also satisfies the innovation constraint, so M sets $\tau^{dual} = b$ and induces $\Delta^{dual} = \bar{\Delta}$.
If $b > \bar{\tau}$, M cannot set $\tau = b$ and preserve high innovation. It compares two feasible policies.

Low fee, high innovation

Set $\tau = \bar{\tau}$ and induce $\Delta = \bar{\Delta}$:

$$\Pi_H = \bar{\tau} G(\nu + \sigma + b).$$

High fee, low innovation

Set $\tau = b$ and accept $\Delta = \Delta^l$:

$$\Pi_L = (b + \max\{\sigma - \Delta^l, 0\}) G(\nu + \Delta^l).$$

Condition (8)

M preserves high innovation when

$$\bar{\tau} G(\nu + \sigma + b) \geq (b + \max\{\sigma - \Delta^l, 0\}) G(\nu + \Delta^l).$$

Otherwise it sets $\tau^{dual} = b$ and accepts $\Delta^{dual} = \Delta^l$.

[Back to Proposition 3](#)

Derivation: Welfare Condition Under Dual-Mode Ban

The welfare-improving case requires $\sigma \leq 0$ and $\Delta^{dual} = \Delta^l$. The ban then replaces low-innovation dual mode with marketplace mode.

Marketplace welfare component

In marketplace mode, the binding price is $p_i^* = b + \Delta^{mkt}$ and S chooses $K'(\Delta^{mkt}) = G(v)$. The relevant surplus generated by platform convenience and innovation is

$$(b + \Delta^{mkt})G(v) - K(\Delta^{mkt}).$$

Low-innovation dual-mode benchmark

With $\Delta^{dual} = \Delta^l$, price squeeze generates transaction value $bG(v + \Delta^l)$ plus consumer gains from the default innovation:

$$\int_0^{v+\Delta^l} [v + \Delta^l - \max\{v, v_o\}] dG(v_o).$$

Condition (9)

The ban raises welfare only if the marketplace component exceeds this low-innovation dual-mode benchmark.

[Back to welfare condition](#)

04

PART 4

Imitation and Self-Preferencing

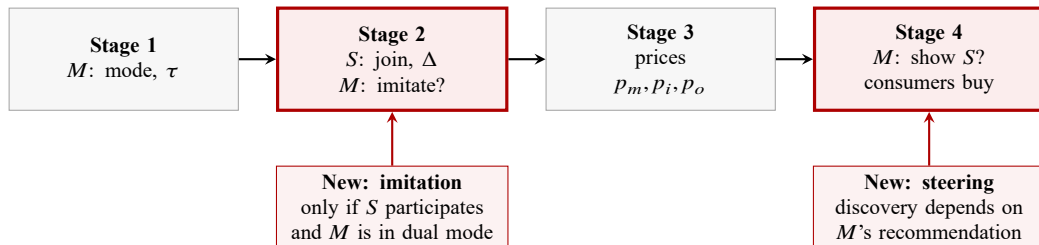
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Timing: Where Section 4 Changes the Game



Changes from the baseline

- M copying choice: M imitates S or not.
- M disclosure choice: M shows S or not.

Review: Price Squeeze Equilibrium

1. Price-squeeze Equilibrium: M shows S , but caps S 's inside price and margin.

- model: M dual mode, shows S .

2. S Price-squeeze price:

$$p_i^* = \max\{\tau, \Delta - \sigma\} \geq \tau \quad (4.3)$$

- participate vs not: $p_i^* - \tau \geq 0 \Rightarrow p_i^* \geq \tau$.
- sell or not: $\Delta - p_i^* \geq \sigma - p_m^* \Rightarrow p_i^* \leq p_m^* + \Delta - \sigma$. (M sets the lowest price (**as seller**) to squeeze S 's)

3. M Price-squeeze price:

$$p_m^* = \max\{\tau - \Delta + \sigma, 0\} \geq 0 \quad (4.4)$$

4. Price-squeeze profits: M get the commission (as marketplace) from S 's sales:

$$\begin{aligned} \Pi^{sqz} &= \Pi^{dual} = \tau G(\nu + b + \Delta - p_i^*) = \tau G(\nu + \sigma + b - p_m^*) \\ \pi^{sqz} &= (p_i^* - \tau)G(\nu + b + \Delta - p_i^*) = (p_i^* - \tau)G(\nu + \sigma + b - p_m^*) \end{aligned} \quad (4.5)$$



Consequence of Self-Preferencing

Self-preferencing: $\tau \leq b \rightarrow \tau \leq b + \Delta$

If

1. Recommendation rule:

$$\Delta + b - p_i \geq \max\{\Delta - p_o, b - \tau, 0\} \quad (4.9)$$

2. High-fee path: M can extract S 's margin as a marketplace.

$$\tau^* = b + \Delta, \quad p_i^* = \tau^*, \quad \pi_S = -K(\Delta) \quad (4.10)$$

3. Hiding path: M shields its own product from competition with S .

$$p_m^* = \min\{\tau, b\} + \sigma \quad (4.11)$$



Consequence of Self-Preferencing

Self-preferencing means that discovery of S is no longer automatic: after observing prices, M can show S or hide it.

Showrooming constraint is relaxed

Without steering, S can use the platform for discovery and then direct consumers to its outside channel, so

$$\tau \leq b. \quad (4.12)$$

With steering, a high-fee price squeeze can still be sustained:

$$\tau = b + \Delta, \quad p_i^* = \tau, \quad \Pi_M = (b + \Delta)G(v), \quad \pi_S = -K(\Delta). \quad (4.13)$$

Competition from S can also be switched off

If M hides S , consumers compare M only with fringe sellers and no purchase. With imitation, this gives

$$p_m^* = b + \Delta, \quad \Pi_M = (b + \Delta)G(v), \quad CS = CS^0. \quad (4.14)$$

Economic meaning: self-preferencing removes the visibility threat that used to discipline M 's commission and pricing. The platform can either tax S through a high fee or steer consumers to its own product, so on-platform price competition largely disappears.

Proposition 5: Dual Mode Equilibrium

Assumptions in the regulated-practice environment

- **Imitation allow:** M can perfectly imitate S 's innovation at no cost, $\sigma = \max\{\Delta, \sigma\}$.
- **Self-preferencing allow:** M can perfectly steer consumers toward its own offer, $\tau = b + \Delta$.
- **Dual mode is feasible:** M both sells its own product and hosts S .

Main variables in equilibrium

- M 's profit: $\Pi^{dual} = \max \{p_m^* G(\nu + b + \Delta^{dual} - p_m^*), \tau^* G(\nu + b + \Delta^{dual} - p_i^*)\} = (b + \max\{\sigma, \Delta^I\})G(\nu)$.
- S 's operating profit: $\pi^{dual} = \max \{(p_i^* - \tau)G(\nu + b + \Delta^{dual} - p_i^*) - K(\Delta^{dual}), 0\} = 0$.
- Innovation level: $\Delta^{dual} = \Delta^I \leftarrow \Delta^{dual} = \arg \max_{\Delta^{dual} \geq \Delta^I} \pi^{dual} - K(\Delta)$.
- Consumer surplus: $CS^{dual} = CS^0 \equiv \int_0^\infty \max\{\nu, \nu_o\} dG(\nu_o) \equiv f(\nu)$.
- Total welfare: $W^{dual} = \Pi^{dual} + \pi^{dual} + CS^{dual}$

Proposition 5: Outcome Vector

Table 1: Dual-mode benchmark with imitation and self-preferencing (Proposition 5)

Case	M 's choice of mode	τ	Π	π	Δ
$\sigma \geq \Delta^l$	Dual - no imitation, M sells	$\tau^* \geq b + \Delta^l$	$(b + \sigma)G(\nu)$	0	Δ^l
$0 < \sigma < \Delta^l$	Dual - high fee or imitation	$\tau^* \geq b$	$(b + \Delta^l)G(\nu)$	0	Δ^l
$\sigma \leq 0$	Dual - high fee or imitation	$\tau^* \geq b$	$(b + \Delta^l)G(\nu)$	0	Δ^l

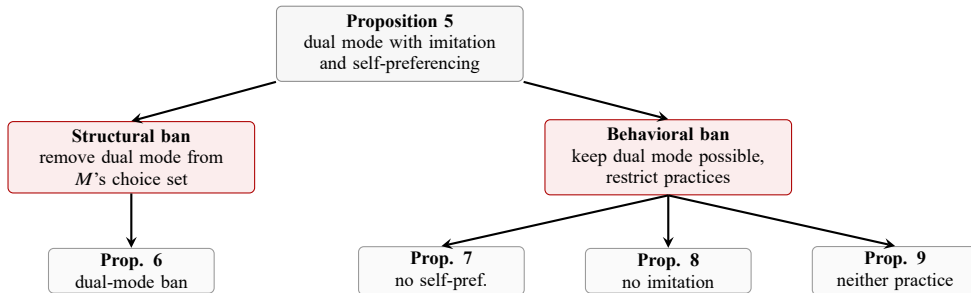
High-fee and imitation paths

- **High fee** (M as marketplace): $\tau = b + \Delta^l$, from self-preferencing M may choose to not show S , thereby capturing S 's full profit.
- **Imitation** (M as seller): $p_m = b + \Delta^l$.
- $\pi = 0$: (1) with a high fee, M extracts all of S 's price margin; (2) with imitation, M makes all sales itself.
- $\Delta = \Delta^l$: since S 's operating profit does not increase with Δ , it has no incentive to innovate above the minimum level.

Policy Map: How the Bans Are Related

Ban types

- **Structural ban:** change M 's business model by removing dual mode; it can shift M to a pure seller or pure marketplace mode.
- **Behavioral ban:** keep dual mode possible but remove specific practices; it can restore price discipline and/or innovation incentives.



Dual-Mode Ban - Setup

Dual-Mode Ban

- **Economic meaning:** M can only choose between being a **seller** or a **marketplace**, but can still choose **imitation** or **self-preference** behavior.
- **Mathematical meaning:** M 's stage-0 choice set is reduced from {Marketplace, Seller, Dual} to {Marketplace, Seller}.

Mode	τ^*	Δ	Π	π	CS
Seller	$\tau^* \geq b$	$\Delta^* = \Delta^l$	$(b + \sigma)G(\nu)$	0	CS^0
Marketplace	$\tau^* \leq b$	$\Delta^* = \Delta^{mkt}$	$\tau^*G(\nu)$	$\Delta^{mkt}G(\nu) - K(\Delta^{mkt})$	CS^0

- $p_m^* = \min\{b, \tau^*\} + \sigma$ and $\tau^* = b$, see Eq. (??).
- $p_i^* = \min\{\Delta^*, b + \Delta^* - \tau^*\}$, see Eq. (??).



Dual-Mode Ban - Equilibrium

Table 2: Effects of a dual-mode ban (Proposition 6) v.s. Table 1

Region	M 's choice of mode	Π	π	$CS = CS^0$	Δ	W
$\sigma \geq \Delta^l$	Seller	$[=]$	0 [=]	$[=]$	Δ^l [=]	$[=]$
$0 < \sigma < \Delta^l$	Seller	$(b + \sigma)G(\nu)$ [↓]	0 [=]	$[=]$	Δ^l [=]	$(b + \sigma)G(\nu) + CS^0$ [↓]
$\sigma \leq 0$	Marketplace	$bG(\nu)$ [↓]	$\Delta^{mkt}G(\nu) - K(\Delta^{mkt})$ [↑]	$[=]$	Δ^{mkt} [↑]	$(b + \Delta^{mkt})G(\nu) - K(\Delta^{mkt}) + CS^0$ [↑]

- $0 < \sigma < \Delta^l$: **(1) math:** $(b + \sigma)G(\nu) < (b + \Delta^l)G(\nu) \rightarrow W \downarrow$. **(2) econ:** M to switch to seller mode, so S 's superior product is no longer discovered through the platform.
- $\sigma \leq 0$: **(1) math:** $bG(\nu) + \Delta^{mkt}G(\nu) - K(\Delta^{mkt}) > bG(\nu) + \Delta^lG(\nu) \rightarrow W \uparrow$. **(2) econ:** M to switch to marketplace mode, so S 's inferior product is now discovered through the platform.

Self-Preferencing Ban - Setup

Self-Preferencing Ban

- **Economic meaning:** if S participates M cannot steer consumers away from S .
- **Mathematical meaning:** the disclosure choice is removed and the showrooming constraint is restored ($\tau^* \leq b$).
- For M : $\tau^* = b$, $p_m^* = b + \max\{\sigma - \Delta^*, 0\}$ (no imitation, imitation)
and $\Pi = \max\{\tau^* G(\nu + b + \Delta^l - p_i^*), p_m^* G(\nu + b + \sigma - p_m^*)\} = (b + \max\{\sigma - \Delta^l, 0\}) G(\nu + \Delta^*)$ (seller, marketplace).
- For S : $\Delta = \Delta^l$, $p_i^* = b$, $p_o^* = 0$
and $\pi = \max\{(p_i^* - \tau^*) G(\nu + b + \Delta^* - p_i^*), p_o^* G(\nu + \Delta^* - p_o^*)\} = 0$ (through M , directly).

Regime	Outcome	τ	Δ	p_m^*	p_s^*	Π	π
$\sigma < \Delta^l$	Dual, imitation, S sells	b	Δ^l	b	$p_i^* = b$	$bG(\nu + \Delta^l)$	0
$\sigma \geq \Delta^l$	Dual, no imitation, M sells	b	Δ^l	$b + \sigma - \Delta^l$	$p_i^* = b$	$(b + \sigma - \Delta^l)G(\nu + \Delta^l)$	0

Self-Preferencing Ban - Equilibrium

Table 3: Effects of banning self-preferencing (Proposition 7) v.s. Table 1

Region	M 's choice of mode	Π	π	CS	Δ	W
$\sigma \geq \Delta^l$	Seller	[=]	[=]	[=]	[=]	[=]
$\underline{\sigma}^{steer} < \sigma < \Delta^l$	Seller	$(b + \sigma)G(\nu)$ [↓]	[=]	[=]	[=]	[↓]
$\sigma \leq \underline{\sigma}^{steer}$	Dual, no self-pref.	$bG(\nu + \Delta^l)$ [↓]	[=]	CS^{ns} [↑]	[=]	$\Pi + CS^{ns}$ [↑]

Note: $CS^{ns} \equiv \int_0^\infty \max\{\nu + \Delta^l, \nu_o\} dG(\nu_o)$; $CS^0 \equiv \int_0^\infty \max\{\nu, \nu_o\} dG(\nu_o)$.

- $\underline{\sigma}^{steer} < \sigma < \Delta^l$: **math**: $(b + \sigma)G(\nu) < (b + \Delta^l)G(\nu)$. **econ**: M switches to seller mode; S is not discovered.
- $\sigma \leq \underline{\sigma}^{steer}$:
 - Π ↓: **math**: $bG(\nu + \Delta^l) < (b + \Delta^l)G(\nu)$ by assumption 2. **econ**: restored competition lowers prices and expands demand.
 - W ↑: **math**: $bG(\nu + \Delta^l) + CS^{ns} > (b + \Delta^l)G(\nu) + CS^0 \Leftrightarrow b[G(\nu + \Delta^l) - G(\nu)] + \int_\nu^{\nu + \Delta^l} (\nu + \Delta^l - \nu_o) dG(\nu_o) > 0$.
econ: consumers are better off with S 's product being discovered, and get the more consumer.

Imitation Ban - Setup

Imitation Ban

- **Economic meaning:** M can credibly commit not to copy S 's product, S again benefit from innovation.
- **Mathematical meaning:** M can only effect high innovation by choosing a fee ($\tau \leq \bar{\tau}$)^a

^a $\bar{\tau}$ is the highest fee that allows S to have high innovation, satisfied $(\bar{\Delta} - \sigma - \bar{\tau})G(\nu + b + \sigma) - K(\bar{\Delta}) = 0$.

Innovation condition (Eq. (4.17) or Eq. (4.18) holds)

- Innovation-preserving condition for M :

$$\bar{\tau}G(\nu + \sigma + b) \geq (b + \max\{\sigma, \Delta^I\})G(\nu). \quad (4.17)$$

left - M 's profit from preserving innovation; right - M 's profit from from rent extraction.

- Non-binding innovation constraint for S :

$$b + \Delta^I \leq \bar{\tau}. \quad (4.18)$$

left - the extractive fee; right - the highest fee compatible with high innovation.



Imitation Ban - Equilibrium

Table 4: Effects of banning imitation (Proposition 8) v.s. Table 1

Region	M 's mode	Π	π	CS	Δ	W
Innovation condition fails	Dual, no imitation	$(b + \max\{\sigma, \Delta^l\})G(\nu)$ [=]	0 [=]	CS^0 [=]	Δ^l [=]	$\Pi + CS^0$ [=]
Innovation condition holds	Dual, no imitation	$\bar{\tau}G(\nu + \sigma + b)$ [↑]	0 [=]	CS^{ni} [↑]	$\bar{\Delta}$ [↑]	$\Pi + CS^{ni}$ [↑]

$$CS^{ni} \equiv \int_0^\infty \max\{\nu + b + \sigma, \nu_o\} dG(\nu_o). \quad CS^0 \equiv \int_0^\infty \max\{\nu, \nu_o\} dG(\nu_o).$$

- Condition fails: **math**: $\tau = b + \Delta^l$ and $\Delta = \Delta^l$, so the outcome equals Proposition 5. **econ**: innovation condition fails.
- Condition holds:
 - Π ↑: **math**: $\bar{\tau}G(\nu + \sigma + b) > (b + \Delta^l)G(\nu)$, by Eq (4.17) or Eq (4.18). **econ**: commitment not to copy lets M induce high innovation and sell to a larger market.
 - W ↑: **math**: Π ↑, CS ↑. **econ**: higher innovation expands demand and consumers benefit.

Policy Comparison: $\sigma > 0$

Table 7: Behavioral remedies relative to a dual-mode ban (Corollary 2)

Region	Remedy	Π	π	CS	Δ	W
$\sigma \in (0, \underline{\sigma}^{steer}]$ $\sigma > \underline{\sigma}^{steer}$	Ban self-preferencing	$\uparrow$	$=$	$\uparrow$	$=$	$\uparrow$
	Ban self-preferencing	$=$	$=$	$=$	$=$	$=$
All $\sigma > 0$	Ban imitation	$\uparrow$	$=$	$\uparrow$	$\uparrow$	$\uparrow$
$\sigma \in (0, \underline{\sigma}]$ $\sigma > \underline{\sigma}$	Ban both practices	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
	Ban both practices	$=$	$=$	$=$	$=$	$=$

Baseline: under a dual-mode ban and $\sigma > 0$, M chooses seller mode.
Arrows are weak changes.

Policy Comparison: $\sigma \leq 0$

Table 8: Behavioral remedies relative to a dual-mode ban (Corollary 2)

Remedy	Π	π	CS	Δ	W
Ban self-preferencing	↑	↓	↑	↓	↑ if condition (9) holds ↓ otherwise
Ban imitation	↑	↓	↑	↑ if Eq. (4.17) holds ↓ otherwise	↑ if Δ increases ↓ otherwise
Ban both practices	↑	↑	↑	↑ if $b \leq \bar{\tau}$ or condition (8) holds ↓ otherwise	↑ if Δ increases or condition (9) holds ↓ otherwise

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PART 5

Policy and Extensions

Presenter: Zhuoyue Wu

Advanced Microeconomics Presentation

Should Platforms Be Allowed to Sell on Their Own Marketplaces?

Hagiu, Teh, and Wright (2022)